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Design of a seismic damper with a structural optimization process

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Abstract. *Several significant seismic events have highlighted the need to assess the seismic vulnerability of critical infrastructures, particularly high-voltage grids, which are crucial for maintaining power supply in emergency scenarios. Despite the installation of advanced nonlinear energy dissipation devices to protect against seismic actions, design documentation is often inadequate, leaving gaps in accurately predicting system behavior. This paper proposes a simplified approach to effectively assess the seismic vulnerability of articulated structures exhibiting nonlinear behavior. A by-pass switch equipped with four wire rope dampers has been subjected to seismic qualification tests at the L.E.D.A. (Laboratory of Earthquake Engineering and Dynamic Analysis) Research Centre of the University of Enna Kore. A Finite Element Method (FEM) model has been developed using displacement and acceleration data, followed by a two-step optimization process using a novel optimisation technique, called Continuous Particle Swarm Optimisation. In the first step, the frequencies and damping were accurately approximated to the experimental values, but the comparison in terms of displacement results remained imprecise. After the optimization of the nonlinear parameters, the dynamic response showed a good degree of agreement with the experimental data in terms of frequency and damping, as well as in terms of displacements and accelerations. This provides a reliable framework for developing simplified and validated FEMs for complex structures with nonlinear dampers.*

1 Introduction

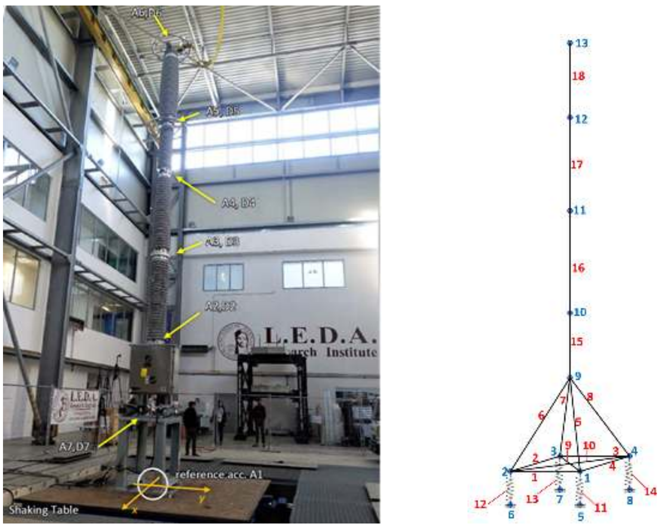
The seismic vulnerability of critical infrastructures, particularly components of high-voltage sub-stations, has received extensive attention following recent earthquakes and extreme weather events. These infrastructures are essential for modern society, as they facilitate the transmission of electricity from generation points to end users. However, their vulnerability to various hazards can lead to severe disruptions, that can have far-reaching economic and social consequences [1, 2]. These events have shown the urgent need to evaluate the seismic vulnerability of complex and highly articulated infrastructures equipped with nonlinear energy dissipation devices, such as wire rope dampers. The use of wire-rope systems has been investigated for seismic protection of equipment within buildings in Ref. [3] and for seismic base-isolation of high-voltage electrical equipment in Ref. [4].These devices, consisting primarily of a single helical stainless steel cable, offer strong vibration isolation capabilities in all three directions due to the cable's mechanical flexibility, while the mutual sliding of the strands provides a high dissipation capacity. In addition, the difference between the design of an artifact and its actual construction creates uncertainty in predicting how the structures will behave mechanically. This uncertainty comes from various factors, such as differences in the size and strength of the materials used and changes in the structural components. Aspects like boundary conditions, manufacturing tolerances, and assembly processes also add to this uncertainty. As a result, accurately predicting the mechanical behaviour of these structures is difficult due to the complex interaction of these many variables.

A high-voltage By-Pass Switch (BPS) with base-mounted wire rope dampers, has been subjected to seismic qualification testing at the L.E.D.A. Research Centre [5], in accordance with the directives of the commissioning authority, as per IEEE Std. 693 [6] Using experimental data on displacement and acceleration, this study proposes a simplified method for assessing seismic vulnerability. The structure has been modelled using a Finite Element Method (FEM) approach and and a two-step optimization procedure has been performed by using the Continuous Particle Swarm Optimization [7, 8]. The nonlinear behaviour of the four wire rope dampers has been modelled by using the rate-independent hysteretic model proposed by Vaiana and Rosati [9].

The following is a summary of the paper's outline. Section 2 introduces the experimental model of the high-voltage BPS, along with its numerical model developed by means of a Matlab® self-developed FEM software. Section 3 provides a detailed explanation of the optimization procedure used to determine the transformer parameters, whereas Section 4 presents the numerical results obtained. Finally, some conclusions are drawn.

2 Experimental setup and Numerical Modelling

The experimental setup consists of a full-scale GL314 By-Pass Switch (BPS), exceeding 10 meters in height, mounted on a support equipped with four galvanized wire rope dampers.



(a) (b)

Figure 1: (a) Experimental setup for shaking table tests; (b) FEM model.

Each damper consists of a single helical stainless steel cable that provides a nonlinear response from the structure even under low seismic intensity conditions. The BPS comprises four hollow ceramic circular elements of varying geometries, linked by metallic flanges, which encase the switching mechanisms, the specifics of which remain unknown. The connection between the BPS and the shaking table has been ensured via a robust steel support structure, consisting of four columns anchored to the shaking table, with a total weight of 295 kg. The structural response has been monitored using accelerometers and reflective spherical markers to track the time histories of accelerations and displacements in the three directions. The experimental configuration, illustrated in Figure1a, shows the structural testing arrangement on one of the shaking tables at the L.E.D.A. Research Centre, highlighting the instrumented points.

Notably, the commissioning entity provided only limited data regarding the geometric and mechanical details of the device, which proved insufficient for the development of a detailed and complex model. Therefore, a simplified concentrated mass model has been developed by means of a Matlab® self-developed FEM software, in which the influence of the steel support structure below the dampers was excluded due to its considerably greater rigidity compared to the BPS structure and the dampers. Therefore, the model focuses on the section of the structure between the dampers and the top of the bypass, with four external rigid constraints applied at nodes 5 to 8, which are connected to four springs simulating the dampers. The connecting frame was modelled as a spatial truss structure consisting of rigid members, as depicted in Figure 1b. The FEM model includes 13 nodes and 18 elements, resulting in a total of 78 degrees of freedom, with external constraints applied only to nodes 5 to 8. Elements 1 to 10 represent the supporting frame mounted on the dampers and have been modelled using rigid members. Elements 11 to 14 are the wire rope dampers, modelled using 3D spring elements with axial and rotational stiffness characteristics, whereas elements 15 to 18 are represented by 3D beam elements, whose damping is modelled by the Rayleigh method as follows:

$$\mathbf{C}_i = \alpha_{0,i}\mathbf{M}_i + \alpha_{1,i}\mathbf{K}_i, \qquad i = 15, 16, 17, 18 \tag{1}$$

The equations of motion of the nonlinear system BPS-dampers are:

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{C}\dot{\mathbf{u}}(t) + +\mathbf{K}\mathbf{u}(t) + \mathbf{F}_{\mathrm{NL}}(t, \mathbf{u}, \dot{\mathbf{u}}) = -\mathbf{M}\boldsymbol{\tau}\ddot{\mathbf{u}}_g^{\mathrm{exp}} \tag{2}$$

where $\ddot{\mathbf{u}}$, $\dot{\mathbf{u}}$ and $\mathbf{u}$ represent the nodal accelerations, velocities, and displacements, respectively, in all 6 degrees of freedom relative to the ground; $\boldsymbol{\tau}$ is the load incidence matrix and $\ddot{\mathbf{u}}_g^{\mathrm{exp}}$ is the vector capturing the experimental acceleration time histories recorded at the base of the structure during the shaking table seismic tests in the x,y and z directions; $\mathbf{F}_{\mathrm{NL}}$ refers to the array containing the nonlinear forces generated by the four wire rope dampers ($i = 11, 12, 13, 14$) in z-direction, modelled by means of Vaiana Rosati Model - Analytical Formulation (VRM AF) [9]:

$$F_{\mathrm{NL},i}(t) = f_e(t) + k_b u(t) + \mathrm{sgn}(\dot{u}(t)) f_0 - \left(f_e(t_p) + k_b u(t_p) + \mathrm{sgn}(\dot{u}(t)) f_0 - f(t_p)\right) e^{-\mathrm{sgn}(\dot{u}(t))\alpha(u(t)-u(t_p))} \tag{3}$$

where k_b is the linear stiffness coefficient, the term $\mathrm{sgn}(\dot{u}(t))$ indicates the sign of the velocity, t_p is the time at the previous state, $f_e(t)$ is the elastic force defined by:

$$f_e(t) = \beta_1 e^{\beta_2 u(t)} - \beta_1 + \frac{4\gamma_1}{1 + e^{-\gamma_2(u(t)-\gamma_3)}} - 2\gamma_1 \tag{4}$$

The parameters f_0, α, β_i ($i = 1, 2$), γ_i ($i = 1, 2, 3$) are constants defining the shape of the hysteretic loop. A classification of all possible shapes is presented in Ref. [9]. Since a thorough analysis of the experimental data did not definitively identify the exact shape of the hysteresis loop, but only allowed an estimation of the maximum force and displacement limits within the hysteresis cycle, the findings of the research carried out by Paolacci and Giannini [4] have been used to guide the shape type selection. Their investigations, which involved cyclic tension-compression tests on three different wire rope models, has shown that all models exhibited hysteresis cycle shapes that were qualitatively similar to the S2.1 shape identified by Vaiana and Rosati [9]. This qualitative agreement supported the choice to adopt the S2.1 shape as the most appropriate representation of the behavior of wire rope dampers under seismic loading. Accordingly, the following conditions have been implemented in the FEM model:

$$\begin{gathered} k_b^+ \neq 0; \quad f_0^+ = f_0^- \neq 0; \quad \alpha^+ = \alpha^- \neq 0; \beta_1^+ = \beta_1^- \neq 0; \quad \beta_2^+ \neq \beta_2^- \neq 0; \\ k_b^- = 0; \quad \gamma_i^+ = \gamma_i^- = 0 \quad (i = 1, 2, 3). \end{gathered} \tag{5}$$

3 Two-step optimisation procedure

In this work, a simplified approach is proposed using a two-step optimization procedure based on a novel technique known as Continuous Particle Swarm Optimization [7]. The mechanical parameters have been optimised in two sequential phases. The first phase focused only on the optimization of linear parameters, referred to as the Linear Optimization (LO) step. The second phase involved the optimization of nonlinear parameters as well, hence this phase is referred to as the Nonlinear Optimization (NLO) step.

For the first optimization step, the parameters to be optimized include the stiffness and damping parameters of the dampers which are assumed to be the same for each damper, along with the elastic modulus and Rayleigh damping coefficients of the ceramic insulators. Thus, the optimization parameters are:

$$\mathbf{x}_{\mathrm{LO}} = [k_z \quad k_x \quad k_{\omega x} \quad c_z \quad c_x \quad c_{\omega x} \quad E \quad \alpha_0 \quad \alpha_1] \tag{6}$$

The aim of this optimization step is to minimize the following objective function concerning the variation of the selected physical parameters:

$$J(\mathbf{x}_{\mathrm{LO}}) = \frac{1}{2}\frac{\|(\mathbf{f}^{\mathrm{num}} - \mathbf{f}^{\mathrm{exp}})\|}{\|(\mathbf{f}^{\mathrm{exp}})\|} + \frac{1}{2}\frac{\|(\boldsymbol{\zeta}^{\mathrm{num}} - \boldsymbol{\zeta}^{\mathrm{exp}})\|}{\|(\boldsymbol{\zeta}^{\mathrm{exp}})\|} \tag{7}$$

in which $\mathbf{f}^{\mathrm{num}}$ and $\mathbf{f}^{\mathrm{exp}}$ represent the numerical and experimental frequency values respectively, and $\boldsymbol{\zeta}^{\mathrm{num}}$ and $\boldsymbol{\zeta}^{\mathrm{exp}}$ correspond to the numerical and experimental damping ratios. Both frequencies and damping ratios are evaluated for the first seven vibration modes only. Here, k_y, c_y and $c_{\omega y}$ are constrained to equal k_x, c_x and $c_{\omega x}$ respectively and the Vaiana Rosati parameters are assumed fixed and calibrated in order to match the experimental estimation of the maximum limits of the force and displacement values observed within the hysteresis cycle. Specifically, the force values range between ± 50,000 N, while displacements are limited to ± 5 cm.

The optimal parameters obtained from the linear optimization phase have subsequently used as initial values for the second optimization, aimed at modeling the nonlinear behavior of the dampers. In this case, the vector of the optimisation parameters are:

$$\mathbf{x}_{\mathrm{NLO}} = [k_b^+ \quad f_0 \quad \alpha \quad \beta_1 \quad \beta_2^+ \quad \beta_2^- \quad k_x \quad k_{\omega x} \quad c_z \quad c_x \quad c_{\omega x} \quad E \quad \alpha_0 \quad \alpha_1] \tag{8}$$

and the objective function to be minimize is in terms of the minimum discrepancy between numerical and experimental displacements in both the x and y directions, particularly concerning nodes 9 to 13.

$$J(\mathbf{x}_{\mathrm{NLO}}) = \sum_i \left\{ w_{ix} \left[\frac{\mathrm{rms}\left(u_{i,x}^{\mathrm{num}}(x, t_i, t_f)\right) - u_{i,x}^{\mathrm{exp}}(t_i, t_f)}{\mathrm{rms}\left(u_{i,x}^{\mathrm{exp}}(t_i, t_f)\right)} \right] + w_{iy} \left[\frac{\mathrm{rms}\left(u_{i,y}^{\mathrm{num}}(x, t_i, t_f)\right) - u_{i,y}^{\mathrm{exp}}(t_i, t_f)}{\mathrm{rms}\left(u_{i,y}^{\mathrm{exp}}(t_i, t_f)\right)} \right] \right\} \tag{9}$$

where w_{ix} and w_{iy} are weighting parameters, $\mathrm{rms}(\cdot)$ is the root mean square function, u_i are the displacements with respect to ground at $i-$th node ($9 \leq i \leq 13$) in x and y global direction, respectively, for the numerical (num) and the experimental (exp) system. The time window (t_i, t_f) have been chosen to be the time interval between the 2% and 98% contribution of the experimental shaking table input acceleration time histories $\ddot{\mathbf{u}}_g^{\mathrm{exp}}$ to the Arias intensity.

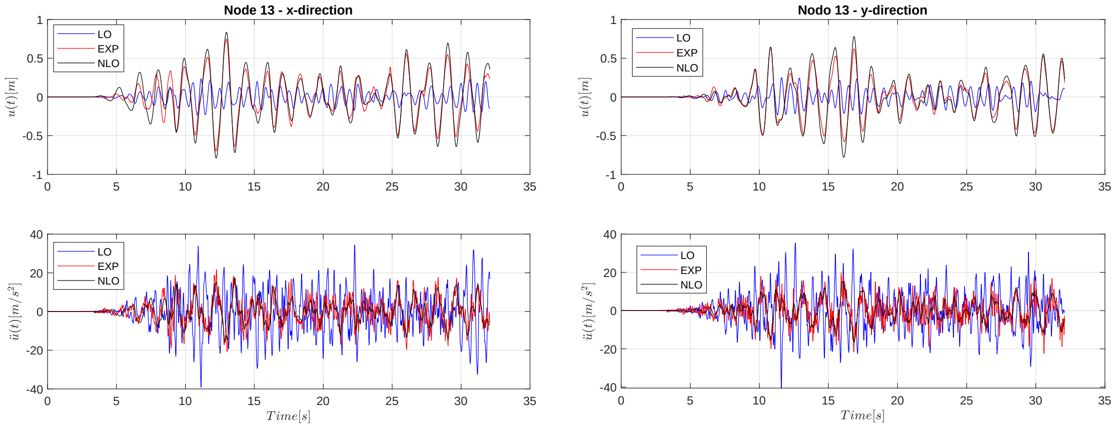
4 Numerical Results

The numerical results obtained from the Linear Optimisation (LO) step and the Nonlinear Optimisation (NLO) step, in terms of frequencies and damping, are listed in Table 1 alongside the experimental values.

Table 1: Comparison between the experimental and numerical frequencies and damping values obtained through the Linear Optimisation (LO) step and the Nonlinear Optimisation (NLO) step

Mode	% mass	direction	f^{exp} [Hz]	$f_{\mathrm{LO}}^{\mathrm{num}}$ [Hz]	$f_{\mathrm{NLO}}^{\mathrm{num}}$ [Hz]	ζ^{exp} [%]	$\zeta_{\mathrm{LO}}^{\mathrm{num}}$ [%]	$\zeta_{\mathrm{NLO}}^{\mathrm{num}}$ [%]
1	98	x	1.30	1.61	1.60	8.26	8.33	8.23
2	98	y	1.30	1.61	1.61	8.44	8.33	8.32
3	62	x	7.28	4.53	4.52	3.76	4.15	4.16
4	62	y	8.01	4.55	4.52	4.53	4.15	4.32
5	35	x	14.77	16.18	16.04	2.63	2.79	3.03
6	35	y	16.23	16.20	16.15	2.97	2.79	2.78
7	99	z	22	21.94	-	8.00	7.99	-

Although the Linear Optimisation (LO) step shows close agreement with the experimental values in terms of frequencies and damping, the numerical response in terms of displacement and acceleration displays significant



(a) (b)

Figure 2: Displacements and accelerations at the top of the structure (node 13) vs time: experimetal results (red line), numerical results by using optimal parameters of Linear Optimisation step (blue line) and Nonlinear Optimisation step (black line) in (a) x-direction; (b) y-direction.

discrepancies when compared to the experimental displacement response, as shown in Figure 2 for example. In the latter, red line represents experimental data while blue line represents numerical simulation by using optimal parameters of Linear Optimisation step.

The results in Table 1 obtained after the Nonlinear Optimisation step indicate that frequencies and damping values exhibit negligible variations compared to those from the Linear Optimisation, with the exception of the seventh mode in the z-direction. During optimization, the axial stiffness of the springs was set to zero, resulting in a notably low numerical frequency in that direction, which was subsequently discarded.

Figure 2 shows a good agreement between the numerical (black line) and experimental data (red line), demonstrating that the proposed methodology effectively defines a set of structural parameters capable of accurately simulating the dynamics of the structure under investigation. Moreover, the numerical hysteresis cycle generated after the second optimisation step demonstrates a more defined shape compared to that generated by the first one (not reported for the sake of brevity), closely approximating the qualitative trend of the cycle proposed by Paolacci and Giannini [4]. Regarding the values derived from the force and displacement intervals, the latter aligns better with experimentally estimated values as shown in Figure 3.

However, the optimization of displacements for nodes 9 and 10, located at the lower part of the BPS, yielded less precise numerical results that are not reported here for the sake of brevity. This discrepancy can be attributed to the seismic test, during which the structure, particularly at the higher points, experienced displacements on the order of several decimeters, exceeding the bounds of small displacements. In contrast, the lower points experienced significantly smaller maximum displacements. In addition, in the objective function defined by Eq. 9, differentiated weights have been assigned to the displacements of various nodes, with the assumption that greater weight is given to higher displacements and lesser weight to those located lower. Consequently, during the second optimization phase, the model exhibited a tendency to align more closely with displacements that have been assigned greater weight. Thus, while the upper points have been approximated with considerable accuracy, the lower regions have not received the same level of precision.

5 Conclusions

This study presents a framework for evaluating the seismic vulnerability of high-voltage By-Pass Switches equipped with four wire rope dampers, employing a two-step optimization approach based on Continuous Particle Swarm Optimization. The results show that while the linear optimization phase closely matched experimental frequencies and damping values, it failed to accurately capture displacement responses. In contrast, the nonlinear optimization phase significantly improved the agreement between numerical and experimental displacements and accelerations. This validation underscores the importance of incorporating nonlinear parameters for accurate structural behavior predictions during seismic events. Additionally, the proposed 2-step optimization approach reduces computational burden compared to a single, larger optimization problem. However, the current model is not always accurate in large displacement scenarios, highlighting the need for a new model to address these cases effectively.

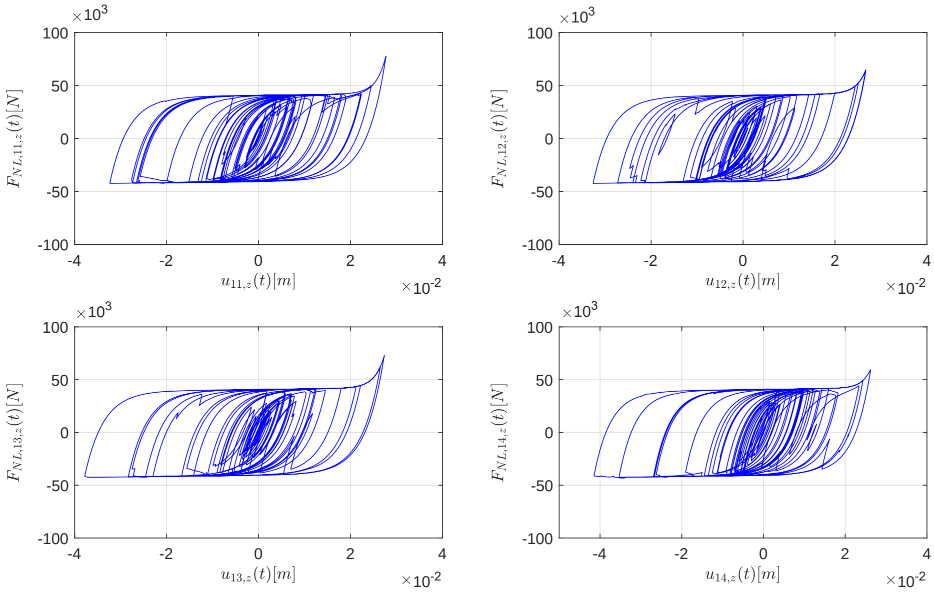


Figure 3: Numerical hysteresis loops of the four dampers obtained after the Nonlinear Optimisation step.

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